To What Extent Is Digital Connectivity Associated with Within-Country Changes in Income Inequality, Controlling for Macroeconomic and Labor Market Conditions?

Carl Von Ossietzky Universität Oldenburg

Faculty of Business Administration, Economics and Law
Chair of Applied Economics

Supervisor:
Lecturer: Prof. Dr. Jürgen Bitzer and Dr. Bernhard Christopher Dannemann

Doctoral Researcher and Teaching Assistant: Abigail Opokua Asare

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Submitted by:

Daniela Borja Polanco Student ID: 6595697

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1. Introduction

Income inequality remains one of the most persistent economic challenges faced by both advanced and developing economies. At the same time, the global economy has experienced a rapid expansion of digital connectivity. Over the past two decades, substantial investments in broadband infrastructure and mobile networks have significantly increased access to the internet, transforming the way individuals access information, participate in labor markets, and engage in economic activity.

This rapid digital expansion has led to the expectation that greater connectivity may promote more inclusive economic development. By providing broader access to information, education, financial services, and employment opportunities, digital technologies have the potential to reduce barriers to economic participation. However, the distributional consequences of digital connectivity remain uncertain. While improved access to digital networks may create new opportunities for individuals and firms, technological change may also disproportionately benefit higher-skilled workers or regions with better infrastructure.

As a result, the relationship between digital connectivity and income inequality remains ambiguous. On the one hand, improved access to digital technologies may expand economic opportunities and reduce inequality[1]. On the other hand, unequal access to digital infrastructure and skills may reinforce existing disparities, particularly across regions and income groups[2].

Despite growing research on digitalization and inequality, important limitations remain. Much of the existing literature relies on cross-country comparisons, making it difficult to isolate the specific effect of digital connectivity from broader structural factors such as income levels, institutions, and labor market conditions. In addition, many analyses rely on composite digitalization indices that combine multiple dimensions of digital transformation, including infrastructure, digital skills, and business adoption. As a result, it is often difficult to identify the independent role of digital connectivity as the underlying infrastructure that enables participation in the digital economy. Furthermore, empirical models do not always explicitly account for labor market mechanisms that may link digitalization and inequality.

The goal of this research is to determine whether the expansion of digital connectivity within countries is associated with changes in income inequality. Specifically, the analysis focuses on internet penetration as a proxy for digital infrastructure. By examining how connectivity evolves within countries over time, this study aims to assess whether improvements in digital access translate into more inclusive economic outcomes.

2. Literature Review

The relationship between digital connectivity and income inequality has received growing attention in empirical economic research. The rapid expansion of internet access, mobile technologies, and digital platforms has transformed production systems, financial markets, and labor markets across the world. However, the direction of the relationship between digital connectivity and within-country income inequality remains theoretically ambiguous and empirically contested.

A substantial strand of the literature finds that digitalization is associated with reductions in income inequality. For example, Tchamyou, Erreygers, and Cassimon provide evidence from African countries that information and communication technologies (ICT) reduce inequality primarily through improvements in financial access[3]. Digital technologies expand both the intensive and extensive margins of financial inclusion, allowing previously excluded populations to participate more actively in the financial system. Similarly, Yin and Choi show that digitalization is associated with lower income inequality in G20 countries, with particularly strong effects in middle-income economies[4]. Using a global panel dataset, Zhu et al. also find that digital service trade is negatively associated with income inequality, suggesting that digital integration into global markets can generate distributional benefits[1].

At the same time, evidence highlight heterogeneous or inequality-increasing effects of digitalization. Nguyen finds that digitalization narrows income inequality in advanced economies but may widen it in developing countries[2]. These results are consistent with the theory of skill-biased technological change, according to which technological progress increases demand for high-skilled labor while potentially displacing low-skilled workers. Similarly, Au shows that the distributive impact of digitalization depends on the specific form of digital development, with different digital indicators displaying varying relationships with inequality[5].

While the existing literature provides important insights, an important limitation remains. Digitalization is often measured using composite digital economy indices that combine multiple dimensions of digital transformation, including infrastructure, digital skills, business integration, and digital trade. Although these measures capture overall digital development, they make it difficult to isolate the specific role of digital connectivity infrastructure in shaping income distribution. Because these indices aggregate several mechanisms, changes in inequality may reflect improvements in skills, business adoption, or digital trade rather than connectivity itself.

To address this limitation, this paper focuses on internet penetration as a direct measure of digital connectivity. Furthermore, although labor market mechanisms are often discussed as potential transmission channels linking digitalization and inequality, empirical specifications frequently rely mainly on macroeconomic controls. The empirical approach therefore incorporates labor market indicators, including unemployment and vulnerable employment, to better capture these dynamics.

3. Data and Descriptive Statistics

3.1. Data Sources and Initial Variable Collection

This paper combines inequality, digital connectivity, and macroeconomic indicators from two primary international databases.

Income inequality and labor market variables are obtained from the World Bank's *World Development Indicators (WDI)*[6]. The primary dependent variable is the Gini index (0–100 scale), which measures the degree of income inequality within a country. Additional inequality-related indicators such as the income share of the top 10 percent, poverty headcount ratios, and mean income of the bottom 40 percent were initially collected to evaluate data coverage and robustness. Labor market controls include total unemployment, youth unemployment, and vulnerable employment.

Digital connectivity indicators are sourced from the International Telecommunication Union (ITU)[7]. These include the percentage of individuals using the Internet, mobile cellular subscriptions per 100 people, and the share of households with Internet access at home.

Macroeconomic controls, GDP per capita, GDP growth, and inflation are also drawn from the World Bank.

The analysis focuses on the period 2005–2024. This restriction reflects improvements in cross-country reporting of ICT indicators beginning in the mid-2000s.

3.2. Variable Coverage and Selection

Before constructing the final panel, data availability was systematically evaluated. For each variable, the proportion of missing observations over the 2005–2024 period was calculated to assess suitability for panel estimation.

The results reveal substantial heterogeneity in coverage across indicators. Several inequality and ICT variables exhibited extremely high levels of missingness exceeding 90 percent of country–year observations in some cases and were therefore excluded to avoid excessive reductions in the estimation sample.

The Gini index, although missing in approximately 73 percent of country–year observations, was retained as the dependent variable due to its central importance in cross-country inequality research and the ability of fixed effects models to accommodate unbalanced panels.

Labor market indicators, including total unemployment, youth unemployment, and vulnerable employment, display comparatively strong coverage (approximately 84 percent availability) and are retained as control variables. Among digital connectivity measures, the percentage of individuals using the Internet and mobile cellular subscriptions per 100 people exhibit strong and consistent reporting, with missingness below 16 percent and 4 percent, respectively. These variables are therefore selected as the main explanatory variables. The share of households with Internet access at home, with moderate coverage, is retained for robustness analysis. Other ICT indicators with sparse or discontinuous coverage are excluded. Macroeconomic controls (GDP growth and inflation) exhibit low levels of missingness and are included to mitigate omitted-variable bias.

Variable selection is guided by three criteria: sufficient temporal and cross-country coverage, theoretical relevance, and adequate within-country variation for fixed effects estimation.

3.3. Panel Construction and Data Preparation

A unified macro-panel dataset is constructed by merging the digital connectivity, inequality, and macroeconomic datasets using ISO3 country codes and calendar year as common identifiers. A left-join strategy is employed, using the connectivity dataset as the base sample to preserve all available observations for the main explanatory variables.

Country identifiers are standardized following the merge, and duplicate country variables created during the joining process are consolidated to ensure consistency.

An external World Bank income classification file is merged into the dataset to incorporate structural country characteristics. Based on this classification, a binary high-income indicator is constructed and treated as time-invariant. This variable is later used to examine heterogeneous effects across development levels.

The analysis period is restricted to 2005–2024. The dataset is further limited to countries with valid ISO3 identifiers and at least ten years of observations. This minimum time threshold ensures sufficient within-country variation for fixed effects estimation while excluding countries with only sporadic reporting.

The resulting macro-panel contains 4,148 country–year observations. The panel is unbalanced, reflecting differences in reporting frequency across countries and variables. This unbalanced structure is deliberately preserved to retain the maximum amount of available information, consistent with empirical macroeconomic panel practice.

In isolated cases where duplicate country–year observations are detected, the observation with the fewest missing values is retained to ensure unique country–year entries while maximizing data completeness.

3.4. Common Estimation Sample and Reproducibility

To ensure strict comparability across regression specifications, a common estimation sample is defined prior to estimation. Observations are retained only if non-missing values are available for:

- The dependent variable (Gini index)
- The main digital connectivity indicators
- Labor market controls
- Macroeconomic controls

This complete-case restriction ensures that differences across model specifications reflect econometric design choices rather than changes in sample composition.

The resulting estimation sample contains 1,404 country–year observations covering 153 countries over the period 2005–2024.

GDP growth and inflation are included to capture short-run macroeconomic fluctuations that may affect income distribution. GDP per capita in levels is excluded from the baseline fixed effects models. Although income level is theoretically relevant, long-run cross-country differences in development are absorbed by country fixed effects, and GDP per capita is highly persistent over

time. Including it alongside fixed effects would introduce redundancy and potential multicollinearity without contributing meaningful identifying variation.

To ensure full reproducibility, the final estimation dataset is exported and stored as a fixed ("frozen") panel file. Because the World Bank database is periodically updated through revisions of historical values and the addition of newly reported observations the number of available country–year observations may change when the script is re-executed in the future. Freezing the estimation sample guarantees that all reported results are based on a stable and replicable dataset.

3.5. Correlation Analysis

Table 1 Presents the Pearson correlation matrix for the main variables in the common estimation sample. The correlations provide preliminary insight into the direction and strength of pairwise relationships, although they do not control for country-specific characteristics or time effects.

Pearson Correlation Matrix

	Gini index (0–100)	Internet use (%)	Unemployment (%)	Youth unemployment (%)	Vulnerable employment (%)	Mobile subscriptions	Internet at home (%)	GDP growth (%)	Inflation (%)
Gini index (0–100)									
Internet use (%)	-0.43								
Unemployment (%)	0.06	-0.01							
Youth unemployment (%)	0.02	0.07	0.94						
Vulnerable employment (%)	0.38	-0.69	-0.15	-0.21					
Mobile subscriptions	-0.26	0.61	0.03	0.12	-0.48				
Internet at home (%)	-0.47	0.95	-0.01	0.08	-0.67	0.59			
GDP growth (%)	0.12	-0.26	-0.12	-0.17	0.25	-0.22	-0.26		
Inflation (%)	0.08	-0.09	-0.02	-0.02	0.11	-0.06	-0.06	-0.02	

Table 1- Pearson Correlation Matrix

Source: Author's own calculations based on World Bank (WDI) and ITU data.

The correlation matrix indicates a moderate negative association between digital connectivity and income inequality. In particular, internet use is negatively correlated with the Gini index (−0.43), suggesting that country–years with higher levels of internet adoption tend to display lower levels of inequality in unconditional terms. Mobile cellular subscriptions also show a negative relationship with inequality, although the magnitude is smaller.

Labor market variables show mixed patterns. Vulnerable employment is positively associated with the Gini index, indicating that structurally fragile labor markets tend to coincide with higher inequality. In contrast, unemployment indicators and macroeconomic variables such as GDP growth and inflation exhibit relatively weak correlations with inequality.

The digital indicators are strongly correlated with each other. Internet use and household internet access display an extremely high correlation (0.95), indicating that both variables capture closely related dimensions of digital development. Including both indicators simultaneously in the regression model could therefore introduce multicollinearity.

To address this issue, internet use is retained as the primary indicator of digital connectivity while excluding household internet access from the regression specifications due to their extremely high correlation. At the same time, mobile cellular subscriptions are preserved as an alternative measure of digital connectivity and are used in additional specifications to assess the robustness of the

results. Similarly, total unemployment is retained as the main indicator of labor market conditions, while youth unemployment is excluded due to its strong correlation with total unemployment.

Another notable pattern is the strong negative relationship between internet connectivity and vulnerable employment (−0.69), suggesting that countries with higher digital access also tend to have more formalized labor markets. This pattern highlights that digital development is closely linked to broader structural transformation processes.

Overall, the correlation analysis suggests a negative unconditional association between digital connectivity and inequality while also revealing strong relationships among digital and labor market variables. These patterns highlight the need for a controlled econometric framework that can distinguish the effects of digital connectivity from broader structural differences across countries. For this reason, the empirical analysis focuses on within-country variation over time using a panel fixed effects approach.

3.6. Descriptive Statistics

Table 2 reports summary statistics for the common estimation sample consisting of 1,404 country–year observations. The variables exhibit substantial cross-country and intertemporal variation, providing a suitable basis for panel estimation.

Summary Statistics

Variable	Mean	Median	SD	Min	Max	N
Gini index (0–100)	36.07	34.60	7.65	23.20	64.80	1404
Internet use (%)	57.67	64.71	28.51	0.22	99.50	1404
Unemployment (%)	7.70	6.38	5.17	0.14	37.32	1404
Vulnerable employment (%)	27.46	21.11	21.05	0.23	90.26	1404
Mobile subscriptions	108.70	114.43	33.63	2.35	214.84	1404
GDP growth (%)	3.12	3.17	4.15	-16.04	27.96	1404
Inflation (%)	5.30	3.09	12.26	-4.45	255.30	1404

Table 2- Descriptive Statistics

Source: Author's own calculations based on World Bank (WDI) and ITU data.

The dependent variable, the Gini index, averages 36.1, indicating moderate income inequality in the sample. The distribution ranges from approximately 23 to 65, revealing considerable heterogeneity between relatively equal and highly unequal societies. The standard deviation of 7.7 suggests meaningful dispersion in inequality levels, while the mean slightly exceeding the median indicates that a subset of high-inequality observations raises the overall average.

Digital connectivity also shows substantial variation across countries and over time. The share of individuals using the Internet averages 57.7 percent, with a wide dispersion and values ranging from very low levels to nearly universal adoption. This variation reflects persistent differences in

digital infrastructure and access across countries. Mobile cellular subscriptions average 108.7 per 100 inhabitants, reflecting widespread multi-SIM ownership in many countries. While internet use captures the extent to which individuals are connected to digital networks, mobile subscriptions provide a complementary indicator of digital engagement, as they typically reflect active participation in mobile communication markets and the willingness to pay for telecommunications services. Although this variable also displays variation, its dispersion is smaller than that of internet use, suggesting that mobile connectivity is more broadly diffused across countries than internet access.

Labor market indicators reveal structural differences across economies. Total unemployment averages 7.7 percent, while vulnerable employment averages 27.5 percent and exhibits particularly large dispersion. The wide range of vulnerable employment highlights substantial differences in labor market structures, from highly formalized economies to contexts characterized by widespread informal or precarious employment.

Macroeconomic variables also display meaningful variation. GDP growth averages 3.1 percent but ranges from deep contractions to rapid expansions, reflecting global and country-specific shocks during the sample period. Inflation averages 5.3 percent but exhibits strong right-skewness, with extreme observations associated with episodes of macroeconomic instability.

Overall, the descriptive statistics reveal three important patterns. First, income inequality varies substantially across countries and over time. Second, digital connectivity indicators show considerable heterogeneity, particularly in internet adoption. Third, labor market structures and macroeconomic conditions differ widely across countries. Figure A- 1 in the Appendix further illustrates these patterns by showing the evolution of internet penetration and income inequality across income groups. While internet adoption expands strongly in high- and upper-middle-income countries, the patterns are more volatile in lower-middle- and low-income economies, where connectivity begins at much lower levels and increases more unevenly over time. At the same time, inequality follows heterogeneous trajectories across country groups. These patterns suggest that simple aggregate trends do not reveal a clear relationship between digital connectivity and income inequality. This motivates the econometric analysis that follows.

4. Empirical strategy

Based on the correlation matrix and the previous variable selection process, the empirical analysis focuses on a limited set of key variables with sufficient data coverage and within-country variation over time. The dependent variable is the Gini index, denoted $Gini_{it}$, where i indexes countries and t indexes years. Internet use (percentage of individuals using the Internet) serves as the main explanatory variable in the baseline specification. Labor market conditions are controlled for using vulnerable employment and total unemployment, while GDP growth and inflation account for short-run macroeconomic fluctuations that may also influence income inequality.

The correlation matrix indicates that digital connectivity indicators are strongly correlated with each other, raising potential multicollinearity concerns if included simultaneously. To avoid this issue, digital variables are not entered together in the same specification. Instead, alternative regressions are estimated in which internet use is replaced by mobile cellular subscriptions per 100 people. This approach allows the robustness of the results to different measures of digital connectivity to be assessed without inflating standard errors due to multicollinearity.

The empirical analysis proceeds in several steps. As a descriptive benchmark, pooled ordinary least squares (OLS) regressions are estimated. The unconditional specification is:

$$Gini_{it} = \beta_0 + \beta_1 Internet_{it} + \varepsilon_{it}$$

which captures the raw cross-country association between digital connectivity and inequality but does not account for structural differences across countries.

A second pooled specification introduces observable labor market and macroeconomic controls:

$$Gini_{it} = \beta_0 + \beta_1 Internet_{it} + \beta_2 Vulnerable_{it} + \beta_3 Unemployment_{it} + \beta_4 GDPgrowth_{it} + \beta_5 Inflation_{it} + \varepsilon_{it}$$

Here, $Vulnerable_{it}$ denotes vulnerable employment, $Unemployment_{it}$ total unemployment, $GDPgrowth_{it}$ real GDP growth, and $Inflation_{it}$ consumer price inflation. These variables capture structural labor market conditions and macroeconomic fluctuations that may influence income distribution. However, pooled OLS estimates remain limited because they ignore unobserved country-specific characteristics.

The main empirical specification therefore employs a two-way fixed effects panel model of the following form:

$$Gini_{it} = \beta_1 Internet_{it} + \beta_2 X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

where X_{it} represents the vector of control variables, α_i captures country fixed effects, and λ_t captures year fixed effects. The country fixed effects control for time-invariant characteristics such as geography, institutional quality, or historical inequality patterns, while year fixed effects account for global shocks affecting all countries simultaneously. This framework isolates within-country variation over time and forms the basis for inference.[8]

Several robustness checks are conducted to assess the stability of the results. First, internet penetration is replaced by mobile cellular subscriptions per 100 people as an alternative measure of digital connectivity:

$$Gini_{it} = \beta_1 Mobile_{it} + \beta_2 X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

Second, to evaluate potential delayed effects and reduce concerns about contemporaneous reverse causality, internet penetration is included with a one-year lag:

$$Gini_{it} = \beta_1 Internet_{i,t-1} + \beta_2 X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

Finally, heterogeneity across development levels is examined by interacting internet penetration with a high-income country indicator:

$$Gini_{it} = \beta_1 Internet_{it} + \beta_2 (Internet_{it} \times HighIncome_i) + \beta_3 X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

In this specification, β_1 represents the effect for non–high-income countries, while $\beta_1 + \beta_2$ captures the total effect for high-income economies.

All specifications are estimated using the common estimation sample to ensure comparability across models.

5. Results

5.1. Descriptive Benchmark: Pooled OLS Results

Table B- 1 (appendix) presents pooled OLS estimates as a descriptive benchmark of the cross-country association between digital connectivity and income inequality. In column (1), which includes only internet use, the coefficient is negative and statistically significant at the 1 percent level. Economically, this suggests that more digitally connected economies are generally characterized by more inclusive income distributions. One possible interpretation is that greater internet access facilitates access to information, financial services, and employment opportunities, which may contribute to reducing income disparities across the population. However, this result reflects cross-country differences and may partly capture broader development patterns rather than a direct effect of connectivity itself.

Column (2) introduces labor market and macroeconomic controls. The coefficient on internet use remains negative and statistically significant but decreases in magnitude to −0.081. This attenuation suggests that part of the raw association is related to differences in labor market structure and macroeconomic conditions across countries. Both vulnerable employment and total unemployment are positively and significantly associated with inequality, indicating that structurally weaker labor markets tend to coincide with higher income dispersion in cross-country comparisons. GDP growth and inflation do not display statistically significant associations in this specification.

While these pooled results illustrate a strong negative cross-country correlation between digital connectivity and inequality, they do not account for unobserved country-specific characteristics. Therefore, they should not be interpreted as evidence of within-country effects. The main inference relies on the fixed effects specifications presented in the following section.

5.2. Main Results: Two-Way Fixed Effects Estimates

Outcome variable: Gini index (0–100)					
	(1) Internet only	(2) + Vulnerable employment	(3) + Unemployment	(4) + GDP growth	(5) + Inflation
Internet use (%)	-0.048***	-0.051***	-0.056***	-0.055***	-0.053***
	(0.007)	(0.008)	(0.008)	(0.008)	(0.008)
Vulnerable employment (%)		-0.052*	-0.074**	-0.072**	-0.070**
		(0.030)	(0.030)	(0.030)	(0.030)
Unemployment (%)			0.140***	0.147***	0.149***
			(0.027)	(0.027)	(0.027)
GDP growth (%)				0.027	0.036*
				(0.021)	(0.021)
Inflation (%)					0.021***
					(0.007)
Num.Obs.	1384	1384	1384	1384	1384
R2 Within Adj.	0.033	0.034	0.054	0.055	0.061

* p < 0.1, ** p < 0.05, *** p < 0.01
Robust standard errors clustered at the country level.
Mobile subscriptions are measured as the number of subscriptions per 100 people.

Table 3- Two-Way Fixed Effects Estimates of Digital Connectivity and Income Inequality

Source: Author's own calculations based on World Bank (WDI) and ITU data.

Table 3 reports the results from the two-way fixed effects specifications, which constitute the main empirical evidence of the paper. By including both country and year fixed effects, these models focus on within-country changes over time rather than differences across countries.

The coefficient on internet use remains negative and statistically significant as additional control variables are introduced. The results indicate that when internet access expands within a country, income inequality tends to decline. This pattern remains visible even after accounting for labor market conditions and macroeconomic fluctuations, suggesting that the association is not simply reflecting broader economic improvements. Economically, the finding is consistent with the idea that greater digital access may widen participation in economic opportunities, reduce information barriers, and improve access to markets and services. Although annual changes in the Gini index are small, inequality typically evolves slowly. Sustained increases in digital connectivity may therefore accumulate into meaningful distributional improvements.

Turning to the control variables, total unemployment shows a positive and highly significant association with inequality across specifications. This implies that worsening labor market conditions within a country are strongly linked to rising income dispersion. Inflation also becomes positive and statistically significant in the full specification, suggesting that macroeconomic

instability may disproportionately affect lower-income groups. GDP growth displays a small positive association with inequality and reaches statistical significance only in the full model. When inflation is introduced, part of the macroeconomic variation previously reflected in growth may be captured by price dynamics, indicating that short-run economic cycles rather than sustained growth are more closely linked to inequality changes in this framework.

Interestingly, vulnerable employment is negatively associated with inequality in the fixed effects models. This contrasts with the positive cross-country correlation observed in the pooled OLS results. The difference highlights that structural differences across countries are not equivalent to short-run changes within countries. Countries with high vulnerable employment tend to be more unequal overall, but when vulnerable employment changes within a country over time, the relationship may follow a different pattern.

Overall, the fixed effects results provide consistent evidence that increases in digital connectivity are associated with modest reductions in income inequality within countries, even after accounting for labor market structure and macroeconomic fluctuations.

5.3. Robustness and Heterogeneous Effects

Table 4 presents additional specifications designed to test the robustness of the main findings and to examine whether the relationship differs across development levels.

Digital Connectivity and Income Inequality (Gini Index)			
	(2) FE (Mobile Subscriptions)	(3) FE (Lagged Internet)	(4) FE (Interaction)
Internet use (%)			-0.052***
			(0.008)
Mobile subscriptions	-0.035***		
	(0.004)		
Internet use (lagged, t−1)		-0.003	
		(0.003)	
Internet use × High-income			0.045***
			(0.008)
Vulnerable employment (%)	-0.073**	-0.020	-0.089***
	(0.029)	(0.029)	(0.030)
Unemployment rate (%)	0.110***	0.132***	0.155***
	(0.027)	(0.028)	(0.027)
GDP growth (%)	0.031	0.054**	0.026
	(0.021)	(0.022)	(0.021)
Inflation (%)	0.024***	0.025***	0.021***
	(0.007)	(0.007)	(0.007)
Num.Obs.	1384	1384	1384
R2 Within Adj.	0.092	0.025	0.085

* p < 0.1, ** p < 0.05, *** p < 0.01
Dependent variable: Gini index (0–100), where higher values indicate greater income inequality.
All specifications include country and year fixed effects.
Standard errors are heteroskedasticity-robust and clustered at the country level.
Lagged internet use refers to internet penetration in year t−1.
In the interaction model, the Internet use coefficient represents the effect in non–high-income countries.
The interaction term captures the additional effect for high-income countries.

Table 4- Robustness Checks and Heterogeneous Effects (Two-Way Fixed Effects)

Source: Author's own calculations based on World Bank (WDI) and ITU data.

First, internet use is replaced by mobile cellular subscriptions as an alternative measure of digital connectivity. The coefficient is negative and statistically significant. This indicates that the association between digital development and lower inequality is not limited to broadband internet

access but also appears when using broader telecommunications penetration. In economic terms, improvements in basic digital infrastructure whether through internet or mobile connectivity seem to coincide with more inclusive income dynamics within countries.

Second, internet use is included with a one-year lag to assess whether the effect operates with a delay. In this specification, the coefficient becomes small and statistically insignificant. This suggests that the relationship between digital connectivity and inequality is primarily contemporaneous. In other words, changes in internet penetration are associated with inequality movements within the same period rather than generating delayed distributional adjustments. The absence of a lagged effect also reduces concerns that the baseline results are driven by slow-moving reverse causality.

Finally, the interaction model examines whether the relationship between digital connectivity and inequality differs across development levels. In the specification with the interaction term, the coefficient on internet use represents the effect for non–high-income countries. The estimate is negative and statistically significant, indicating that increases in internet penetration are associated with reductions in inequality in these economies. The interaction term between internet use and the high-income indicator is positive and statistically significant. This implies that the effect of internet expansion is substantially smaller in high-income countries. Specifically, combining the baseline coefficient and the interaction term suggests that the overall effect in high-income economies is close to zero. Economically, this result indicates that digital connectivity may have stronger distributional effects in countries where access to digital infrastructure is still expanding, while additional increases in internet penetration in already highly connected economies generate only limited changes in income distribution.

Across all robustness specifications, unemployment continues to show a strong positive association with inequality, highlighting the central role of labor market conditions in shaping income dispersion. Inflation also remains positively related to inequality, suggesting that macroeconomic instability may disproportionately affect lower-income groups.

Overall, the robustness checks confirm that the negative association between digital connectivity and inequality is stable across alternative measures and model variations. At the same time, the heterogeneity results indicate that the relationship is not uniform across development levels and appears more pronounced in countries where digital expansion remains transformative.

6. Conclusion

This paper examines whether increases in digital connectivity within countries are associated with changes in income inequality. Using a macro-panel of 153 countries over the period 2005–2024 and a two-way fixed effects framework, the analysis focused on within-country variation while controlling for labor market conditions and macroeconomic fluctuations.

The results indicate that higher internet penetration is associated with modest reductions in income inequality within countries. This relationship remains robust when alternative measures of digital connectivity, such as mobile cellular subscriptions, are used. However, the effect appears to be primarily contemporaneous, as lagged internet penetration does not show a statistically significant association with inequality.

The findings also reveal important heterogeneity across development levels. The inequality-reducing relationship between digital connectivity and inequality is stronger in non–high-income countries, while the effect is economically negligible in high-income economies. This suggests that digital expansion may have greater distributional relevance in contexts where access to digital infrastructure is still expanding.

At the same time, labor market conditions remain a key determinant of income inequality. Across specifications, higher unemployment is consistently associated with greater income dispersion, highlighting the continued importance of labor market dynamics in shaping income distribution.

Overall, the findings suggest that digital connectivity is linked to inequality dynamics within countries, particularly in developing and middle-income contexts. While the results should not be interpreted as strictly causal, the within-country panel framework provides systematic evidence that digital expansion is associated with changes in income distribution over time.

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Appendix A – Exploratory Patterns in Digital Connectivity and Inequality

Figure A- 1 Illustrates the evolution of internet penetration and income inequality across income groups. The top panels show internet use for high-income and upper-middle-income countries (left) and for lower-middle-income and low-income countries (right). The bottom panels display the corresponding trends in the Gini index.

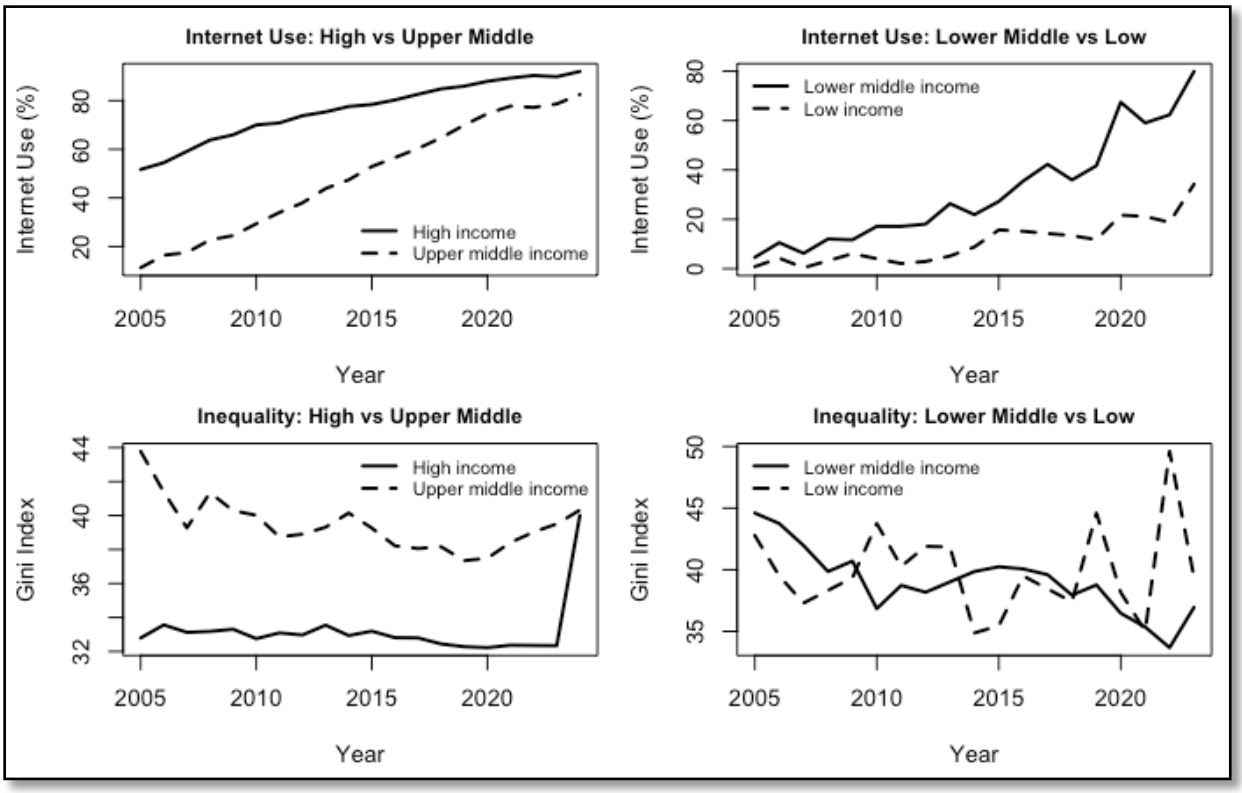

Figure A- 1 Internet Penetration and Income Inequality by Income Group (2005–2024)

Source: Author's own calculations based on World Bank (WDI) and ITU data.

Appendix B – Additional Regression Results

This appendix reports additional regression results used as descriptive benchmarks. Table B- 1 Table B- 1Pooled OLS Estimates of Digital Connectivity and Income Inequalitypresents pooled OLS estimates of the relationship between digital connectivity and income inequality before introducing country and year fixed effects. These specifications capture cross-country associations and are reported for comparison with the fixed effects models presented in the main text.

Outcome variable: Gini index (0–100)		
	(1) Pooled OLS	(2) Pooled OLS + Controls
Internet use (%)	-0.118***	-0.081***
	(0.007)	(0.009)
Vulnerable employment (%)		0.070***
		(0.012)
Unemployment (%)		0.124***
		(0.037)
GDP growth (%)		0.036
		(0.047)
Inflation (%)		0.020
		(0.015)
Num.Obs.	1384	1384
R2	0.191	0.214
R2 Adj.	0.190	0.211

* p < 0.1, ** p < 0.05, *** p < 0.01
Dependent variable: Gini index (0–100).
Standard errors are heteroskedasticity-robust and clustered at the country level.
This specification does not include country or year fixed effects.

Table B- 1Pooled OLS Estimates of Digital Connectivity and Income Inequality

Source: Author's own calculations based on World Bank (WDI) and ITU data.